

VIETNAM NATIONAL UNIVERSITY, HANOI
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REPORT OF FINAL EXAM PROJECT

**Topic: Riding the Green Wave: Social-Psychological
Drivers of Investment Intentions Toward Sustainable
Cryptocurrencies Among Vietnamese Investors in
Vietnam in 2025**

Subject: **Research methods for Natural Sciences and Technology**

Class: **INS1076 - INS107604**

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I. Introduction

1. Background of the study

Over the past decade, the rapid growth of digital assets has coincided with increasing attention to environmental sustainability. According to Chainalysis (2024), global cryptocurrency users exceeded 560 million in 2024, while Vietnam remained one of the world's leading adoption markets, with approximately 21.2% of adults owning cryptocurrency despite the absence of a comprehensive regulatory framework. At the same time, concerns regarding blockchain energy consumption have intensified. The Cambridge Centre for Alternative Finance (CCAF, 2024) estimated that Bitcoin mining consumed 141.4 TWh of electricity annually, comparable to the energy usage of some mid-sized countries. In response, the industry has increasingly shifted toward energy-efficient consensus mechanisms. Notably, Ethereum's Merge reduced network energy consumption by 99.95% (Ethereum Foundation, 2023), while platforms such as Algorand, Solana, Cardano, and Hedera have positioned themselves as carbon-neutral or carbon-negative sustainable cryptocurrencies.

Concurrently, ESG-oriented investing has expanded rapidly, with global sustainable investment assets under management reaching USD 30.3 trillion (GSIA, 2024). Younger investors have shown the strongest preference for sustainable financial products (Deloitte, 2024). This trend is particularly relevant in Vietnam, where investors under the age of 35 accounted for approximately 64% of newly opened securities accounts in 2023 (HoSE, 2024). Together, these developments highlight Vietnam as an important yet underexplored context for examining the determinants of sustainable cryptocurrency investment intention.

2. Statement of the problem

Despite the dramatic growth of sustainable blockchain projects and the documented enthusiasm of Vietnamese retail investors for digital assets, the behavioral determinants of green cryptocurrency investment intention remain poorly understood within the Vietnamese socio-cultural and regulatory context. Most extant research focuses on conventional digital assets, predominantly in developed economies, and does not adequately account for the psychological dynamics distinguishing green investment decisions from general cryptocurrency adoption.

A particularly underexplored problem concerns greenwashing perception — the belief held by investors that environmental claims by blockchain projects are primarily marketing constructs rather than genuine commitments. In the cryptocurrency sector, where technical complexity is high and independent verification of environmental claims is difficult, greenwashing is a structurally persistent risk. Vietnamese retail investors, the overwhelming majority of whom are individual, non-institutional participants with limited technical expertise, are especially vulnerable to deceptive green marketing. This creates a distinctive trust problem: even investors who are genuinely environmentally motivated may hesitate to commit capital to sustainable cryptocurrencies if they perceive green claims as unverifiable or manipulative. Neither green trust nor greenwashing perception has been formally incorporated into the Vietnamese cryptocurrency investment literature.

3. Purpose and significance of the study

This study investigates the socio-psychological factors determining Vietnamese investors' behavioral intention to invest in sustainable cryptocurrencies, integrating the Theory of Planned Behavior (TPB) and Norm Activation Theory (NAT) within a nine-hypothesis model that explicitly incorporates greenwashing perception and green trust. Academically, the study extends both theoretical frameworks into an underexplored behavioral domain, provides the first systematic empirical evidence from the Vietnamese sustainable cryptocurrency context using a 7-point Likert scale, and introduces fsQCA as a methodological tool capable of identifying equifinal causal configurations that symmetric approaches cannot detect.

Practically, the findings bear direct relevance for Vietnam's State Securities Commission and State Bank of Vietnam — specifically for designing regulatory sandboxes and investor protection frameworks. Sustainable blockchain developers may use the findings to communicate environmental credentials more credibly. FinTech platforms can align product offerings with the psychological drivers identified. Investors themselves may benefit from the study's implicit contribution to financial literacy by making explicit the socio-psychological factors that shape and may distort their decision-making.

4. Research objectives

The general objective is to examine the socio-psychological determinants of Vietnamese investors' behavioral intention to invest in sustainable cryptocurrencies in 2025, using an integrated TPB–NAT framework analyzed through PLS-SEM and fsQCA. Specific objectives are:

- To examine the direct effect of environmental concern on sustainable cryptocurrency investment intention.
- To assess the influence of social influence on investment intention.
- To determine the relationship between perceived financial benefit and investment intention.
- To evaluate the negative effect of perceived risk on investment intention.
- To investigate the direct positive role of green trust on investment intention.
- To analyze the influence of government support on investment intention.
- To examine the negative direct effect of greenwashing perception on investment intention.
- To assess the negative moderating role of greenwashing perception on the green trust–investment intention relationship.
- To test the mediating role of green trust between environmental concern and investment intention.
- To identify, via fsQCA, configurational combinations of antecedent conditions sufficient for high investment intention.

II. Literature Review

1. Overview of the field

1.1.Sustainable finance and ESG investing

Sustainable finance integrates Environmental, Social, and Governance (ESG) criteria into financial decision-making. Reinforced by the Paris Agreement and UN Sustainable Development Goals 7, 13, and 17, approximately 90% of reviewed studies report a nonnegative relationship between ESG criteria and corporate financial performance (Friede, Busch, & Bassen, 2015). ESG assets under management surpassed USD 30.3 trillion globally (GSIA, 2024), reflecting the mainstreaming of responsible investment across institutional and retail sectors.

1.2.Sustainable cryptocurrencies: definition and taxonomy

For this study, sustainable cryptocurrencies refer to digital assets that minimize environmental impact through energy-efficient consensus mechanisms such as Proof-of-Stake, carbon offsetting, or renewable energy integration. This definition explicitly excludes energy-intensive Proof-of-Work currencies such as Bitcoin. Key examples include Ethereum (post-Merge), Cardano (ADA), Algorand (ALGO), Solana (SOL), Tezos (XTZ),

and Hedera Hashgraph (HBAR). This definitional clarity is critical in the Vietnamese context, where respondents may conflate all cryptocurrencies with Bitcoin's high energy consumption, potentially confounding perceptions of environmental concern and green trust.

1.3.The greenwashing problem in blockchain

Greenwashing — where blockchain projects exaggerate or misrepresent environmental performance to attract ESG-oriented investors — represents a structurally persistent challenge. Park and Kim (2020) found that misleading sustainability claims significantly reduce investor trust once exposed. A 2023 survey by the Blockchain Transparency Institute reported that more than 40% of projects promoting environmental credentials had not released independently verifiable sustainability reports, underscoring the empirical basis for treating greenwashing perception as a psychologically salient construct among retail investors.

2. Previous studies and findings related to the topic

2.1.Theoretical foundations

Ajzen's (1991) Theory of Planned Behavior posits that behavioral intention the strongest proximate predictor of actual behavior is a function of attitude toward the behavior, subjective norms (social influence), and perceived behavioral control. Fishbein and Ajzen (2010) documented that TPB models typically explain 40–65% of variance in behavioral intention. In the cryptocurrency-specific literature, Albayati, Kim, and Rho (2020) applied TPB to a Jordanian sample of 283 participants, finding that attitude ($\beta = 0.41$, $p < .001$) and subjective norms ($\beta = 0.29$, $p < .01$) were significant predictors of Bitcoin acceptance intention. Nguyen, Nguyen, and Nguyen (2021), in the Vietnamese context specifically, found that subjective norms explained 18.7% of variance in cryptocurrency investment intention ($\beta = 0.33$, $p < .01$) among 310 millennial investors. The present study extends TPB by incorporating financial and institutional factors perceived financial benefit, perceived risk, and government support relevant to sustainable cryptocurrency investment, alongside the core TPB component of social influence.

Schwartz's (1977) Norm Activation Theory (NAT) provides a complementary moral-normative framework. NAT proposes that personal moral norms are activated when individuals develop awareness of the harmful consequences of inaction (awareness of consequences) and ascribe personal responsibility for those consequences (ascription of responsibility). Applied to investment behavior, NAT predicts that investors who are aware of

conventional cryptocurrency's environmental harms and feel personally responsible for addressing them will redirect investment intention toward green alternatives. Environmental concern serves as a proxy indicator representing this norm activation process, consistent with its established use in sustainable investment research (Liu, Zhao, & Li, 2022). Glac (2009) demonstrated that moral norm activation significantly predicted socially responsible investment participation among U.S. retail investors. The integration of NAT with TPB enables comprehensive coverage of both economic rationality dimensions and moral-normative dimensions of sustainable cryptocurrency investment.

2.2. Empirical evidence on key constructs

Environmental concern, defined as an individual's awareness of and concern about environmental degradation, has consistently been identified as a key driver of pro-environmental behavior. Liu et al. (2022) found that environmental awareness was the strongest predictor of green fund purchase intention among 412 Chinese investors ($\beta = 0.46$, $p < .001$), explaining 61.3% of behavioral variance. Similarly, Hsu et al. (2017) reported a significant positive effect on green product purchase intention among Thai consumers ($\beta = 0.38$, $p < .001$). These findings suggest that environmentally concerned investors are more likely to support sustainable cryptocurrency investments.

Social influence refers to the impact of peers, social networks, and subjective norms on individual decision-making. Arias-Oliva et al. (2019), using a sample of 417 Spanish users, found that social influence positively affected cryptocurrency adoption intention ($\beta = 0.27$, $p < .05$). Given Vietnam's collectivist cultural orientation (Hofstede, 1980), peer recommendations and community discussions within cryptocurrency groups are expected to further strengthen investment intention.

Investment decisions are generally based on the evaluation of expected gains and potential losses. Conti et al. (2018), in a review of 126 cryptocurrency studies, identified expected financial return as the most frequently reported investment motive. Conversely, Kim et al. (2020), studying 405 South Korean investors, found that perceived risk significantly weakened investment intention ($\beta = -0.21$, $p < .01$). Consistent with Prospect Theory (Kahneman & Tversky, 1979), investors tend to weigh potential losses more heavily than gains, suggesting a positive effect of perceived financial benefit and a negative effect of perceived risk on sustainable cryptocurrency investment intention.

Green trust reflects investors' confidence in the environmental credibility of a product or investment. Chen (2010) validated the green trust construct with strong reliability (Cronbach's $\alpha = 0.89$; AVE = 0.64). Hur et al. (2014) further demonstrated that green trust mediated the relationship between green CSR and purchase intention, accounting for approximately 34% of the total effect. In sustainable cryptocurrency markets, investors who trust environmental claims are expected to exhibit stronger investment intentions.

Government support, including regulatory clarity, policy incentives, and institutional endorsement, is an important determinant of investment behavior. Shen et al. (2021) found that perceived government support was the strongest predictor of FinTech adoption intention among 368 Chinese users ($\beta = 0.34$, $p < .001$). Similarly, Kaur and Arora (2021) reported a significant positive effect on sustainable investment intention among Indian investors ($\beta = 0.29$, $p < .05$). Therefore, increasing regulatory support in Vietnam is expected to enhance confidence in sustainable cryptocurrency investments.

Greenwashing perception refers to the belief that environmental claims are misleading or exaggerated. Siano et al. (2017) found that perceived greenwashing significantly reduced green purchase intention among 412 Italian consumers ($\beta = -0.31$, $p < .001$). Likewise, Chen and Chang (2013) showed that greenwashing perception negatively moderated the relationship between green trust and purchase intention ($\beta = -0.18$, $p < .05$). Therefore, investors who perceive sustainable cryptocurrency projects as engaging in greenwashing are expected to demonstrate lower investment intention and weaker trust-based investment behavior.

Author(s) & Year	Context / Sample	Key Constructs	Method	Key Finding(s)	Relevance
Albayati et al. (2020)	Jordan; n=283; crypto users	Attitude, SN, PBC → Intention (TPB)	PLS-SEM	Attitude $\beta=0.41^{***}$, SN $\beta=0.29^{**}$	TPB baseline for crypto

Nguyen et al. (2021)	Vietnam; n=310; millennials	SN, attitude, PBC → crypto intention	SEM	SN explains 18.7% of variance	Vietnamese crypto context
Liu et al. (2022)	China; n=412; retail investors	Environmental awareness, green trust → green fund intention	PLS-SEM	Env. awareness $\beta=0.46^{***}$; $R^2=61.3\%$	Green investment NAT evidence
Riedl & Smeets (2017)	Netherlands; n=2,117; panel	Environmental values → SRI fund ownership	Logit regression	Env. values > financial expectations	Moral motivation for SRI
Arias-Oliva et al. (2019)	Spain; n=417; crypto users	Social influence → crypto use intention	SEM	SI $\beta=0.27^*$	Social norm in crypto
Kim et al. (2020)	South Korea; n=405; investors	Return expectations, perceived risk → intention	Regression	Risk moderates return–intention; $\beta=-0.21^{**}$	Prospect theory in crypto
Shen et al. (2021)	China; n=368; FinTech users	Government support → FinTech adoption	PLS-SEM	Govt. support $\beta=0.34^{***}$	Institutional trust driver
Chen & Chang (2013)	Taiwan; consumers	Greenwashing, green trust → green purchase	SEM	Greenwashing moderates GT→PI; $\beta=-0.18^*$	Greenwashing moderation
Siano et al. (2017)	Italy; n=412; consumers	Greenwashing perception → green purchase intention	Regression	GP $\beta=-0.31^{***}$	Greenwashing direct effect

Hur et al. (2014)	Korea; n=302; consumers	Green CSR → brand equity (GT mediates)	PLS-SEM	Indirect effect = 34% of total	Green trust mediation
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Table 1. Summary of selected empirical studies on cryptocurrency adoption and sustainable investment intention

3. Gaps in the literature and the need for the proposed study

Four interrelated gaps collectively justify the proposed study. Theoretically, while TPB and NAT have each been applied separately to cryptocurrency adoption and green consumption, no published study has formally integrated both frameworks to explain sustainable cryptocurrency investment intention. The synthesis proposed here — linking NAT's norm-activation mechanism with TPB's rational-calculus dimensions — constitutes a meaningful theoretical contribution. Contextually, the Vietnamese sustainable cryptocurrency investment landscape is virtually unrepresented in the empirical literature, despite Vietnam's remarkable crypto engagement profile and its distinctive regulatory environment. Methodologically, the exclusive reliance on symmetric analytical methods in prior studies precludes detection of equifinal causal configurations; the combination of PLS-SEM and fsQCA directly addresses this gap. Conceptually, greenwashing perception — while empirically documented in green consumption contexts — has not been formally modeled as a determinant of sustainable cryptocurrency investment intention in any published study, representing a construct-level gap of direct practical relevance to the Vietnamese retail investor context.

4. Theoretical framework and conceptual framework

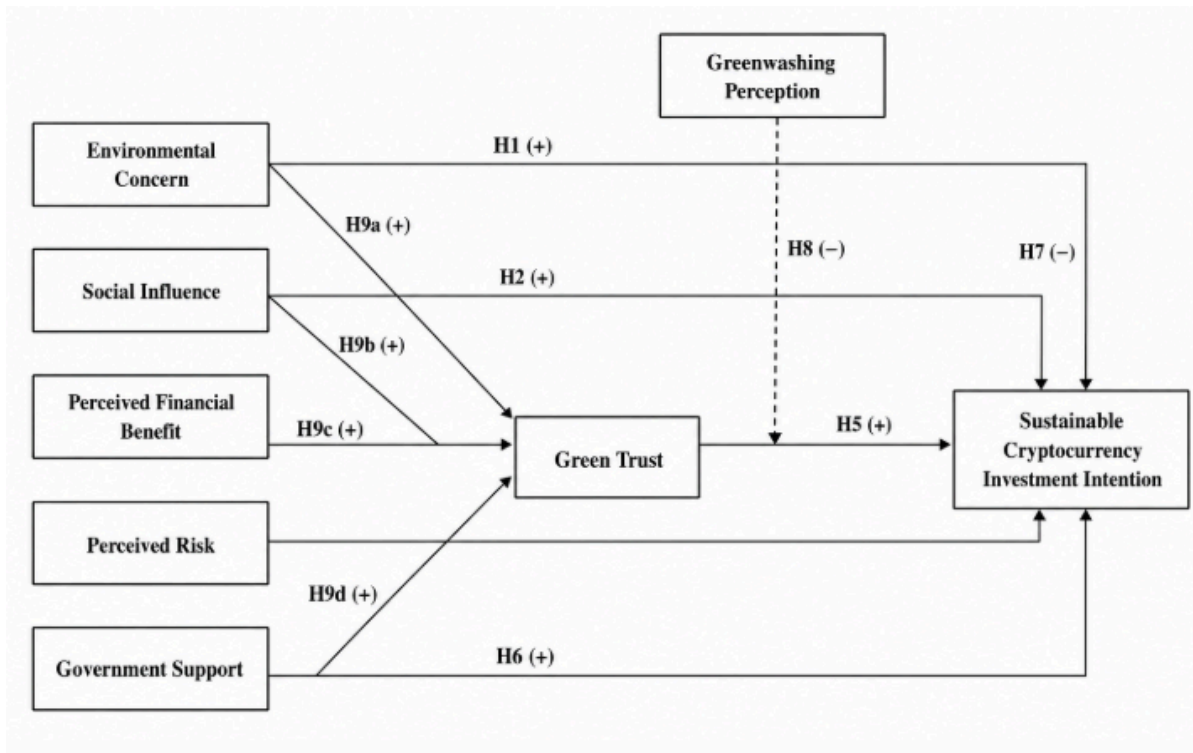


Figure 1. Conceptual Framework: Socio-psychological determinants of sustainable cryptocurrency investment intention

Nine directional hypotheses are formally stated as follows.

- H1: Environmental concern positively influences sustainable cryptocurrency investment intention among Vietnamese investors.
- H2: Social influence positively influences sustainable cryptocurrency investment intention.
- H3: Perceived financial benefit positively influences sustainable cryptocurrency investment intention.
- H4: Perceived risk negatively influences sustainable cryptocurrency investment intention.
- H5: Green trust positively influences sustainable cryptocurrency investment intention.
- H6: Government support positively influences sustainable cryptocurrency investment intention.
- H7: Greenwashing perception negatively influences sustainable cryptocurrency investment intention.

- H8: Greenwashing perception negatively moderates the relationship between green trust and investment intention, such that the positive effect of green trust becomes significantly weaker when greenwashing perception is high.
- H9a: Green trust positively mediates the relationship between environmental concern and sustainable cryptocurrency investment intention.
- H9b: Green trust positively mediates the relationship between social influence and sustainable cryptocurrency investment intention.
- H9c: Green trust positively mediates the relationship between perceived financial benefit and sustainable cryptocurrency investment intention.
- H9d: Green trust positively mediates the relationship between government support and sustainable cryptocurrency investment intention.

III. Methodology

1. Research design

This study adopts a quantitative, cross-sectional, explanatory research design within the positivist tradition to test theoretically derived hypotheses and establish relationships among latent constructs. The analytical strategy integrates PLS-SEM (SmartPLS 4) to evaluate the symmetric hypotheses H1–H9d alongside fsQCA (fsQCA 4.1) to identify configurations of conditions sufficient for high investment intention (RQ2). This mixed-methodological approach optimizes the explanation of complex behavioral phenomena.

The target population consists of adult Vietnamese individuals in 2025 who possess baseline cryptocurrency awareness and investment interest. Utilizing a non-probability purposive and snowball sampling strategy, respondents will be recruited from Vietnamese cryptocurrency communities on Telegram, Facebook, Discord, and Reddit. Two preliminary screening questions regarding awareness of green cryptocurrencies and recent market activity will filter out non-target participants to mitigate self-selection bias.

Sample size determination followed multiple criteria. The 10× rule (Hair, Risher, Sarstedt, & Ringle, 2019) yields a minimum of 70 (10×7 , the maximum number of arrows entering any construct). G*Power 3.1 analysis for linear multiple regression with 7 predictors ($f^2 = 0.15$, $\alpha = .05$, power = .80) yields a minimum of 103. The minimum recommended for stable mediation indirect effects is 200 (Baron & Kenny, 1986). After adjusting for 20%

non-response attrition and 15% invalid response rate, the adopted target is 400 usable responses, requiring distribution of approximately 590 surveys.

2. Data collection methods and instruments

Data will be collected via a structured self-administered online questionnaire distributed through Google Forms. The instrument employs a 7-point Likert response scale (1 = Strongly Disagree; 7 = Strongly Agree) for all reflective construct items. Preston and Colman (2000) found that 7-point scales produced the highest combination of reliability, validity, discriminating power, and orthogonality, making them optimal for capturing nuanced attitudinal dimensions characteristic of sustainable finance behavior.

All items are adapted from previously validated instruments in the sustainable finance, cryptocurrency, and green marketing literatures, with lexical modifications to ensure contextual specificity to sustainable cryptocurrency investment. The instrument will be written in Vietnamese and subjected to professional back-translation into English by a bilingual finance academic to verify construct equivalence (Brislin, 1970). A two-stage pilot test will be conducted: Stage 1 involves cognitive interviewing with five academic experts to assess face validity and item clarity; Stage 2 involves a pilot survey with 40 target population participants to assess item reliability (target Cronbach's $\alpha \geq .70$ per construct).

Construct	Measurement Items (7-point Likert)	Source(s)	# Items
Environmental Concern (EC)	EC1: I am seriously concerned about the state of the environment.	Dunlap et al. (2000); Liu et al. (2022)	4
	EC2: Cryptocurrency energy consumption is a serious environmental threat.		
	EC3: I believe it is important to invest in environmentally sustainable projects.		

	EC4: Environmental degradation motivates my investment decisions.		
Social Influence (SI)	SI1: People important to me think I should invest in sustainable crypto.	Venkatesh et al. (2003); Arias-Oliva et al. (2019)	4
	SI2: My peers actively discuss and invest in green digital assets.		
	SI3: Cryptocurrency influencers I follow promote sustainable projects.		
	SI4: Community endorsement matters to my investment decisions.		
Perceived Financial Benefit (PFB)	PFB1: Sustainable cryptocurrencies offer competitive financial returns.	Conti et al. (2018); Kahneman & Tversky (1979)	4
	PFB2: Green crypto assets are likely to increase in value long-term.		
	PFB3: ESG-aligned crypto projects attract institutional investment.		
	PFB4: I expect sustainable crypto to outperform conventional crypto.		

Perceived Risk (PR)	PR1: I perceive investing in sustainable cryptocurrencies as risky.	Featherman & Pavlou (2003); Kim et al. (2020)	4
	PR2: The volatility of green crypto assets concerns me.		
	PR3: Regulatory uncertainty increases my perceived investment risk.		
	PR4: I am uncertain about the financial safety of green crypto assets.		
Green Trust (GT)	GT1: I trust that sustainable crypto projects genuinely reduce environmental impact.	Chen (2010); Hur et al. (2014)	4
	GT2: Sustainable blockchain platforms reliably deliver on their green promises.		
	GT3: I believe environmental claims by green crypto projects are credible.		
	GT4: I am confident that green consensus mechanisms are as effective as claimed.		
Government Support (GS)	GS1: Government policies in Vietnam support sustainable crypto investment.	Shen et al. (2021); Kaur & Arora (2021)	4

	GS2: Regulatory clarity encourages me to invest in green digital assets.		
	GS3: I trust government oversight to protect sustainable crypto investors.		
	GS4: Policy incentives make sustainable crypto investment more attractive.		
Greenwashing Perception (GP)	GP1: I believe many crypto projects exaggerate their environmental credentials.	Siano et al. (2017); Chen & Chang (2013)	4
	GP2: Green marketing claims by blockchain firms are often misleading.		
	GP3: It is difficult to verify the true environmental impact of crypto projects.		
	GP4: I am skeptical of sustainability reports published by crypto firms.		
Investment Intention (II) — Dependent Variable	II1: I intend to invest in sustainable cryptocurrencies within the next 12 months.	Ajzen (1991); Nguyen et al. (2021)	4
	II2: I plan to allocate a portion of my portfolio to green digital assets.		

	II3: I would recommend sustainable crypto investment to peers.		
	II4: I am willing to actively research sustainable cryptocurrency options.		

Table 2. Measurement items, sources, and 7-point Likert anchors

Ethical oversight will be maintained through the home institution's research ethics review protocol. Informed consent will be obtained digitally before participants access the instrument, with a clear statement of the study's purpose, voluntary nature, anonymity guarantees, and data use policy. No personally identifiable information will be collected. Data will be stored on password-protected encrypted servers accessible only to the research team.

3. Data analysis

Data analysis will be conducted using SPSS 29, SmartPLS 4, and fsQCA 4.1. Given that all constructs are collected through a single self-report questionnaire, common method bias (CMB) will be assessed using Harman's single-factor test (single factor < 50% of total variance) and full-collinearity VIF values ($VIF < 3.3$).

Hypotheses H1–H9d will be tested using PLS-SEM following the two-step approach. The measurement model will be evaluated through indicator reliability (outer loadings ≥ 0.70), internal consistency reliability (Cronbach's α and composite reliability ≥ 0.70), convergent validity (AVE ≥ 0.50), discriminant validity (HTMT < 0.85), and multicollinearity ($VIF < 5.0$). The structural model will be assessed using bootstrapping with 5,000 resamples to estimate path coefficients and significance levels ($p < .05$). Model explanatory and predictive power will be examined through R^2 , Q^2 (> 0), and effect sizes f^2 (0.02, 0.15, and 0.35 indicating small, medium, and large effects, respectively). Mediation hypotheses (H9a–H9d) will be evaluated using bootstrapped indirect effects with 95% bias-corrected confidence intervals, while moderation (H8) will be tested using the two-stage interaction approach in SmartPLS 4.

PLS-SEM construct scores will be calibrated into fuzzy sets using the direct calibration method with thresholds of 0.95 (full membership), 0.50 (crossover point), and 0.05 (full non-membership). Necessary conditions will be assessed using consistency and coverage values above 0.90. Subsequently, a truth table will be generated using a frequency threshold of $n \geq 4$ and a consistency threshold of 0.80. Boolean minimization will then be applied to derive parsimonious, intermediate, and complex solutions, with the intermediate solution reported as the primary result. This analysis enables the examination of equifinality and causal asymmetry in sustainable cryptocurrency investment behavior.

IV. Expected Results

The measurement model is anticipated to satisfy all established psychometric thresholds: Cronbach's $\alpha \geq 0.79$ and composite reliability ≥ 0.84 for all constructs; AVE ≥ 0.50 , confirming convergent validity; and HTMT ratios below 0.85, confirming discriminant validity. Full collinearity VIFs are expected to remain below 3.3, indicating the absence of problematic CMB.

For the structural model, all nine hypotheses are anticipated to be supported. The strongest positive direct effects are expected for perceived financial benefit (H3: $\beta = 0.30\text{--}0.46$, $p < .001$) and environmental concern (H1: $\beta = 0.25\text{--}0.40$, $p < .05$), consistent with Liu et al. (2022) and prospect theory predictions. Green trust (H5: $\beta = 0.22\text{--}0.36$) and social influence (H2: $\beta = 0.18\text{--}0.32$) are expected to demonstrate significant positive effects. Government support (H6: $\beta = 0.15\text{--}0.28$) and perceived risk (H4: $\beta = -0.15$ to -0.28) are expected to produce smaller but significant positive and negative effects, respectively. Greenwashing perception is expected to exert a significant negative direct effect on investment intention (H7: $\beta = -0.18$ to -0.30), and to significantly negatively moderate the green trust–investment intention relationship (H8: interaction $\beta = -0.10$ to -0.20). The complex mediation multi-path outlined (H9a, H9b, H9c, H9d) is expected to yield significant positive indirect effects : $\beta = 0.08\text{--}0.16$, 95% CI non-overlapping zero), consistent with the partial mediation pattern documented by Hur et al. (2014) in analogous green consumption contexts. The combined model is anticipated to explain 55–65% of variance in investment intention (R^2).

H	Predictor	Outcome	Expected β (95% CI)	p-value	f²	Supported?
H1	Environmental Concern	Investment Intention	0.25–0.40	< .05	Medium	Supported
H2	Social Influence	Investment Intention	0.18–0.32	< .05	Small–Med	Supported
H3	Perceived Financial Benefit	Investment Intention	0.30–0.46	< .001	Large	Supported
H4	Perceived Risk	Investment Intention	–0.15 – –0.28	< .05	Small	Supported
H5	Green Trust	Investment Intention	0.22–0.36	< .01	Medium	Supported
H6	Government Support	Investment Intention	0.15–0.28	< .05	Small	Supported
H7	Greenwashing Perception	Investment Intention	–0.18 – –0.30	< .01	Small–Med	Supported
H8 (Mod)	GP $\times$ Green Trust	Investment Intention	–0.10 – –0.20	< .05	Small	Supported
H9a (Med)	Environmental Concern via Green Trust	Investment Intention	0.08–0.16 (indirect)	< .05	Not applicable	Supported
H9b (Med)	Social Influence via Green Trust	Investment Intention	0.06 – 0.14 (indirect)	< .05	Not applicable	Supported
H9c (Med)	Financial Benefit via Green Trust	Investment Intention	0.07 – 0.15 (indirect)	< .05	Not applicable	Supported

H9d (Med)	Government Support via Green Trust	Investment Intention	0.05 – 0.12 (indirect)	< .05	Not applicable	Supported
Overall R²			55–65% of variance in Investment Intention explained			

Table 3. Anticipated structural model path coefficients and hypothesis testing results (H1–H9d)

The fsQCA analysis for RQ2 is expected to reveal four alternative pathways leading to high sustainable cryptocurrency investment intention, supporting the principle of equifinality.

- Configuration 1 (Eco-Trusting Investor) combines high environmental concern and high green trust with low perceived risk and low greenwashing perception as core conditions, representing environmentally motivated investors who trust sustainability claims.
- Configuration 2 (Financial Pragmatist) is characterized by high perceived financial benefit and low perceived risk as core conditions, indicating investors whose decisions are primarily driven by expected economic returns.
- Configuration 3 (Policy-Reliant Skeptic) centers on high government support, high green trust, and low greenwashing perception, reflecting investors who rely on regulatory endorsement and institutional credibility.
- Configuration 4 (Socially Embedded Investor) is anchored by high social influence, supported by green trust and government support, representing investors whose intentions are strongly shaped by community and peer networks.

All configurations are expected to achieve consistency scores above 0.80, indicating sufficient pathways to sustainable cryptocurrency investment intention.

V. Budget and Timeline

1. Estimated costs for conducting the study

#	Budget Item	Cost (VND)	Notes
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1	SmartPLS 4 – Student academic license (12 months)	1,500,000	Essential for PLS-SEM analysis
2	Pilot study incentives (40 respondents × VND 30,000)	1,200,000	Motivate pilot participation
3	Database & article access (~15 Scopus/WoS articles)	1,200,000	Full-text journal access
4	Vietnamese–English back-translation service	700,000	Bilingual finance academic
5	Internet & cloud storage (6 months)	600,000	Data security & sharing
6	Expert cognitive interviews (5 experts — travel & catering)	500,000	Face validity assessment
7	Printing & binding (3 bound final copies)	450,000	Institutional submission
8	Miscellaneous contingency	550,000	Unforeseen expenses
—	fsQCA 4.1 & G*Power 3.1	0	Open-source, no cost
TOTAL		6,700,000 VND	≈ USD 270

Table 4. Estimated research budget (Vietnamese Dong, VND)

2. Funding sources

The study will be self-funded by the primary researcher, with supplementary support from the home institution's student research grant scheme. No external commercial funding sources will be sought, ensuring independence of the research findings and eliminating conflicts of interest in the reporting of results.

3. Timeline for completing the study

Research Phase / Task	M1	M2	M3	M4	M5	M6	M7	M8	M9
Proposal development & literature review	•	•							
Ethics approval submission	★								
Instrument design & back-translation			•	★					
Expert cognitive interview (pilot stage 1)			•						
Pilot survey (n=40) & refinement				★					
Full-scale data collection (~590 surveys)					•	★			
Data cleaning, screening & CMB testing							•		

PLS-SEM analysis (SmartPLS 4)							•	•	
fsQCA analysis (fsQCA 4.1)								•	★
Report writing & revision								•	★
Peer review & final submission									★
Key Milestones	Ethics approval			Instrument finalized		Data collection closed			Final submission

Table 5. Proposed research timeline — 9 months (Gantt chart format)

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